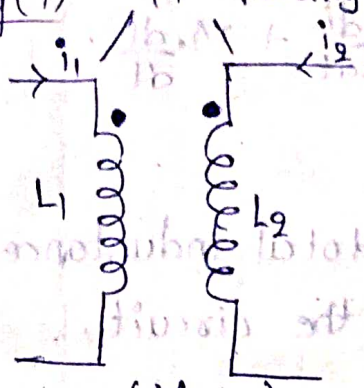
24/1/11

DOT CONVENTION:

1. Dot convention is to find the sign of mutual inductance (M).
2. The sign of mutual induced emf is as follows.

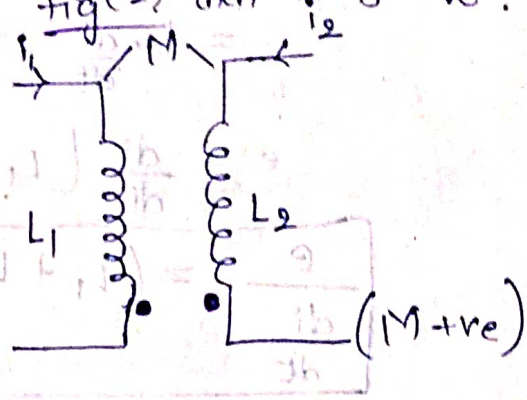
Rules: 1: When both the currents passing through the dot then 'M' is +ve.
2: If one current is passing through the dot & the 2nd current is passing out of dot, then 'M' is -ve.

fig(1)



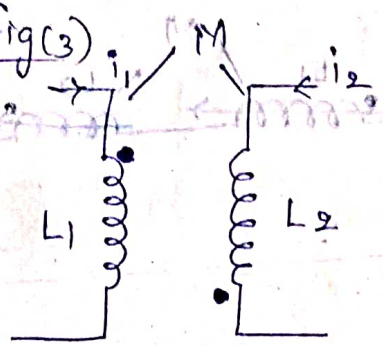
(M+ve)

fig(2)



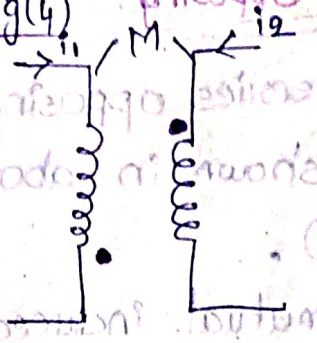
(M+ve)

fig(3)



(M-ve)

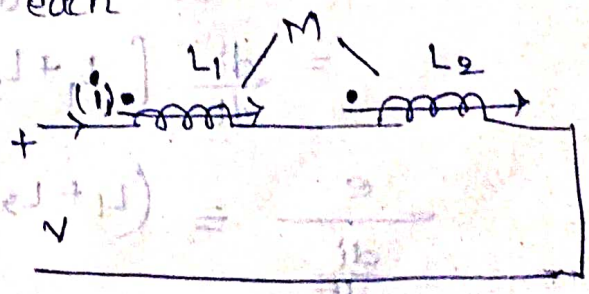
fig(4)



(M-ve)

★★★ Imp 1. Derive an expression for the equivalent inductance of the two coils when it is connected in (i) Series aiding (ii) Series opposing.

A. Series aiding: For "series aiding" connection, the total emf induced in each of the coils L_1 and L_2, is due to the coils of self inductance. The circuit is as shown in the above fig(a).



The emf induced in coil 1 is $e_1 = L_1 \cdot \frac{di}{dt} + M \cdot \frac{di}{dt}$

||ly $e_2 = L_2 \cdot \frac{di}{dt} + M \cdot \frac{di}{dt}$

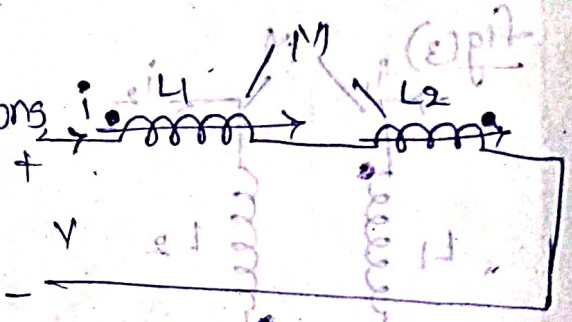
∴ Total induced emf in the circuit is given by $e = e_1 + e_2$

$$= L_1 \cdot \frac{di}{dt} + M \cdot \frac{di}{dt} + L_2 \cdot \frac{di}{dt} + M \cdot \frac{di}{dt}$$

$$e = \frac{di}{dt}(L_1 + L_2 + 2M)$$

$$\boxed{\frac{e}{\frac{di}{dt}} = (L_1 + L_2 + 2M)}$$ total inductance of the circuit.

Series Opposing :-



For series opposing connections are shown in above fig (b).

The mutual induced emf opposes the self induced emf

so, $e_1 = L_1 \cdot \frac{di}{dt} - M \cdot \frac{di}{dt}$

$e_2 = L_2 \cdot \frac{di}{dt} - M \cdot \frac{di}{dt}$

Total induced emf, $e = e_1 + e_2$

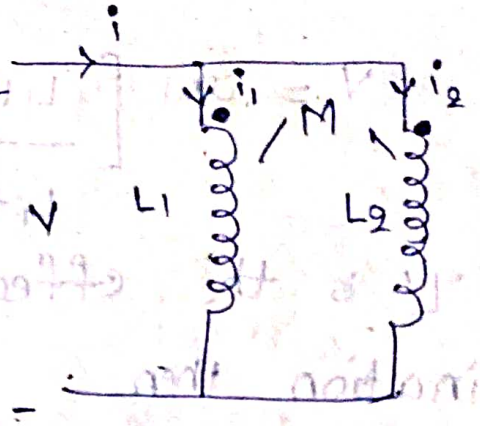
$$e = L_1 \cdot \frac{di}{dt} - M \cdot \frac{di}{dt} + L \cdot \frac{di}{dt} - M \cdot \frac{di}{dt}$$

$$= \frac{di}{dt}(L_1 + L_2 - 2M)$$

$$\boxed{\frac{e}{\frac{di}{dt}} = (L_1 + L_2 - 2M)}$$ total inductance of the circuit

2. Derive an expression for the equivalent inductance of two coils in parallel with
i) Parallel aiding ii) Parallel opposing.

Paralled aiding:-



Consider a parallel coupling of two inductance as shown in above fig(a)

Applying KVL to both of the loops we get.

1st loop: $J\omega L_1 \cdot I_1 + J\omega M i_2 = V \rightarrow (1)$

2nd loop: $J\omega L_2 i_2 + J\omega M i_1 = V \rightarrow (2)$

$$J\omega L_1 \cdot i_1 + J\omega M i_2 = J\omega L_2 i_2 + J\omega M i_1$$

$$i = i_1 + i_2$$

$$i - i_1 = i_2$$

$$J\omega L_1 i_1 + J\omega M(i - i_1) = J\omega L_2(i - i_1) + J\omega M i_1$$

$$J\omega L_1 i_1 + J\omega M i - J\omega i_1 = J\omega L_2 i - J\omega L_2 i_1 + J\omega M i_1$$

$$J\omega L_1 i_1 + J\omega L_2 i_1 - J\omega M i_1 - J\omega M i_1 = J\omega L_2 i - J\omega M i$$

$$\cancel{J\omega} i_1 (L_1 + L_2 - M - M) = \cancel{J\omega} i (L_2 - M)$$

$$i_1 = \frac{i(L_2 - M)}{L_1 + L_2 - 2M}$$

$$i_2 = \frac{i(L_1 - M)}{L_1 + L_2 - 2M}$$

substituting the values of i_1 & i_2 in eq. (1) we get

$$v = j\omega L_1\left[\frac{i(L_2-m)}{L_1+L_2-2m}\right] + j\omega M\left[\frac{i(L_1-m)}{L_1+L_2-2m}\right]$$

$$= j\omega i\left[\frac{L_1(L_2-m)}{L_1+L_2-2m}\right] + j\omega i\left[\frac{m(L_1-m)}{L_1+L_2-2m}\right]$$

$$= j\omega i\left[\frac{L_1(L_2-m)}{L_1+L_2-2m} + \frac{m(L_1-m)}{L_1+L_2-2m}\right]$$

$$= j\omega i\left[\frac{L_1(L_2-m)+m(L_1-m)}{L_1+L_2-2m}\right]$$

$$= j\omega i\left[\frac{L_1L_2-\cancel{L_1m}+\cancel{mL_1}-m^2}{L_1+L_2-2m}\right]$$

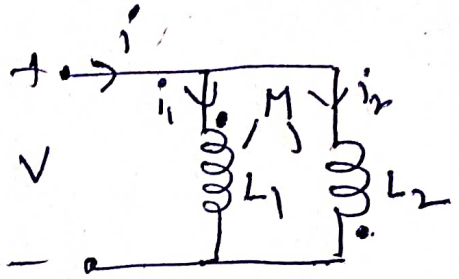
$$v = j\omega i\left[\frac{L_1L_2-m^2}{L_1+L_2-2m}\right] \longrightarrow (3)$$

If L is the effective inductance of parallel combination then $v = j\omega L_{eff}\, i \rightarrow (4)$

comparing (3) & (4)

we have $L_{eff} = \dfrac{L_1L_2-m^2}{L_1+L_2-2m}$

Parallel opposing



consider a parallel coupling of two inductance as shown in the above fig (a).

Applying KVL to both of the loops we get.

First loop :- $j\omega L_1 i_1 - j\omega m i_2 = V \rightarrow (1)$

second loop : $j\omega L_2 i_2 - j\omega m i_1 = 0 \rightarrow (2)$

Equating equation (1) & (2)

$$j\omega L_1 i_1 - j\omega m i_2 = j\omega L_2 i_2 - j\omega m i_1$$

w.k.t $i = i_1 + i_2$

$i_2 = i - i_1$

$$j\omega L_1 i_1 - j\omega m(i - i_1) = j\omega L_2 (i - i_1) - j\omega m i_1$$

$$j\omega L_1 i_1 - j\omega m i + j\omega m i_1 = j\omega L_2 i - j\omega L_2 i_1 - j\omega m i_1$$

$$j\omega L_1 i_1 + j\omega m i_1 + j\omega L_2 i_1 + j\omega m i_1 = j\omega L_2 i + j\omega m i$$

$$\cancel{j\omega} i_1 [L_1 + m + L_2 + m] = \cancel{j\omega} i (L_2 + m)$$

$$i_1 = \frac{i(L_2 + m)}{L_1 + L_2 + 2m}$$

similarly $i_2 = \dfrac{i(L_1 + M)}{L_1 + L_2 + 2m}$

substituting the values of i_1 & i_2 in equation (1)

we get

$$V = j\omega L_1 \left[\frac{i(L_2+M)}{L_1+L_2+2M} \right] - j\omega M \left[\frac{i(L_1+M)}{L_1+L_2+2M} \right]$$

$$= j\omega i \left[\frac{L_1(L_2+M)}{L_1+L_2+2M} \right] - j\omega i \left[\frac{M(L_1+M)}{L_1+L_2+2M} \right]$$

$$= j\omega i \left[\frac{L_1(L_2+M)}{L_1+L_2+2M} - \frac{M(L_1+M)}{L_1+L_2+2M} \right]$$

$$= j\omega i \left[\frac{L_1(L_2+M) - M(L_1+M)}{L_1+L_2+2M} \right]$$

$$= j\omega i \left[\frac{L_1L_2 + \cancel{L_1M} - \cancel{ML_1} - M^2}{L_1+L_2+2M} \right]$$

$$V = j\omega i \left[\frac{L_1L_2 - M^2}{L_1+L_2+2M} \right] \longrightarrow (3)$$

If L is the effective inductance of parallel combination

$$V = j\omega L_{eff}\, i \longrightarrow (4)$$

comparing eq (3) & (4) we have

$$L_{eff} = \frac{L_1L_2 - M^2}{L_1+L_2+2M}$$